

Asymptotic enumeration of tied football seasons

2 October 2026

Abstract

For a double round-robin league with labeled teams, distinguished matches, and the usual 3/1/0 scoring rule, let $a(n)$ count seasons in which every team has the same score. We prove a leading equivalent for OEIS A380592, including its nonconstant three-residue theta factor. The proof uses mean-matching exponential tilts and a sharp global Fourier localization: a negative quartic gives an exact local Gaussian majorant, and a separate outlier argument controls its complement. The limiting common score is a discrete Gaussian whose center differs from its mean. We then verify the hypotheses of Isaev's complex cumulant theorem to obtain an expansion to every fixed inverse-power order, give a finite exact coefficient algorithm, compute the first correction, and formulate residue-aware inverse threshold bounds. The supplied replay checks the finite algebra; the analytic estimates and their scope are proved in the text.

Contents

1	Model and main results	2
2	Mean matching and covariance	4
3	Sharp global localization	4
4	The local integral and the score shift	6
5	Dominated summation and the conditional common score	8
6	Expansion to every fixed order	8
7	The first correction	12
8	Residue-aware asymptotic inversion	14
9	Prior work and scope	15
10	Reproducibility and limits	15

1 Model and main results

There are n labeled teams. Every unordered pair plays two distinguished matches. A match between i and j has score outcomes $(3, 0)$, $(1, 1)$, and $(0, 3)$. Put $M = n(n-1)$, the number of matches. Then

$$a(n) = \sum_{k=2(n-1)}^{3(n-1)} [x_1^k \cdots x_n^k] \prod_{i < j} (x_i^3 + x_i x_j + x_j^3)^2. \quad (1.1)$$

The endpoints follow by summing all scores: each match contributes either 2 or 3. This counts neither unlabeled leagues nor score tables alone. If outcomes are independent and uniform and X_i denotes team i 's score, then $a(n) = 3^M \mathbb{P}(X_1 = \cdots = X_n)$.

The exact enumeration paper of Jehn, Habermann and Lavrov [1] studies this model and gives values through $n = 8$. The values in OEIS A380592 [2] are

$$1, 3, 27, 1083, 296081, 696779523, 16503494334993, 3439079361325736243.$$

Define the positive, real, 1-periodic function

$$\Theta(c) = \sum_{j \in \mathbb{Z}} e^{-4\pi^2 j^2 / 9} e^{2\pi i j c}, \quad (1.2)$$

$$= \frac{3}{2\sqrt{\pi}} \sum_{k \in \mathbb{Z}} e^{-9(k-c)^2 / 4}, \quad (1.3)$$

where the second equality is Poisson summation. Write

$$\mu_n = \frac{8(n-1)}{3}, \quad c_n = \mu_n - \frac{13}{84}, \quad B_n = 3^M n^{-1/2} \left(\frac{9}{56\pi n} \right)^{(n-1)/2} e^{-9829/65856}. \quad (1.4)$$

Theorem 1.1 (Leading equivalent). *As $n \rightarrow \infty$,*

$$a(n) \sim B_n \Theta(c_n). \quad (1.5)$$

Equivalently, retaining the exact transverse covariance,

$$a(n) \sim 3^M (n-1)^{-1/2} \left(\frac{9}{2\pi(28n-2)} \right)^{(n-1)/2} e^{-12181/65856} \Theta(c_n). \quad (1.6)$$

Because $c_{n+3} - c_n = 8$, the multiplier depends only on $n \bmod 3$. It cannot be replaced by 1 in an asymptotic equivalent.

Theorem 1.2 (Every fixed order). *For every fixed integer $R \geq 0$, there are explicitly computable 1-periodic functions $H_0, \dots, H_R$ such that*

$$a(n) = B_n \left(\sum_{j=0}^R \frac{H_j(c_n)}{n^j} + O_R(n^{-R-1}) \right), \quad H_0 = \Theta. \quad (1.7)$$

Each H_j is a finite rational linear combination of derivatives of Θ . In particular, set

$$Q(\delta) = \frac{22180735}{45177216} + \frac{98841}{153664} \delta - \frac{5283}{3136} \delta^2 - \frac{9}{8} \delta^3. \quad (1.8)$$

For $r \in \{0, 1, 2\}$ and any $n \equiv r \pmod{3}$, define

$$R_r = \frac{\sum_{k \in \mathbb{Z}} e^{-9(k-c_n)^2 / 4} Q(k - \mu_n)}{\sum_{k \in \mathbb{Z}} e^{-9(k-c_n)^2 / 4}}. \quad (1.9)$$

Then

$$a(n) = B_n \Theta(c_n) \left(1 + \frac{R_{n \bmod 3}}{n} + O(n^{-2}) \right). \quad (1.10)$$

The defining sums are exact. The following rounded decimals are numerical diagnostics, not interval-certified enclosures.

$n \bmod 3$	$c_n \pmod{1}$	$\Theta(c_n)$	$R_{n \bmod 3}$
0	5/28	1.01079874779311	0.04288551652736
1	-13/84	1.01402025457159	0.19984197501234
2	43/84	0.97518099763530	0.04601392878474

Sections 2–5 give an elementary proof of Theorem 1.1. Sections 6–7 prove Theorem 1.2, invoking a stated complex-cumulant theorem with verified hypotheses. Section 8 treats inversion.

[write] **Added 2 October 2026, batch 77.** This report is a single manuscript, printed in full: manuscript 67 of batch 77 of ProveIt’s incoming-reports intake, delivered as the archive `tied-football-report.zip` (wrapper directory `tied-football-release/`, main file `report/tied-football-asymptotics.tex`, now `article.tex`) in commit `096ee7b87`, and placed in commit `d0e6008d9` in the research-report collection, among the OEIS-sequence asymptotics reports. The manuscript names no ProveIt commit and describes no search of the repository, so it has no pin; its author line is empty. The bounded search of Section 9 is the manuscript’s own literature search; the intake’s untruncated search of the repository (the research-report collection, the transseries volumes and `0eis/`) found no other treatment of A380592 or of tied seasons. The nearest batch-77 neighbour, the collection report `a000571-tournament-score-sequences` (batch-77 manuscript 68), counts score sequences of tournaments from their exact generating function; it is a different model and does not use the tilted Fourier localization of this report, and neither report uses the other. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. The writing step prefixed every label with `tfs:`, labelled Section 1, added these [write] notes and two references [8, 9], and set the bibliography ragged-right; no statement, proof or number of the manuscript was altered.

Notation. Each symbol is printed as delivered; several letters carry two meanings in different sections:

symbol	meanings (and what each is <i>not</i>)
Q, Q_0	the first-correction polynomial $Q(\delta)$ of (1.8); Q_0 is a small closed interval of tilt parameters q about $1/3$ (Section 3), not a value of Q .
R, R_r	a truncation order in Theorem 1.2 and Section 8; R_r ($r = n \bmod 3$) is the first-correction constant of (1.9).
B_n, B, b_3, B_4	the carrier B_n of (1.4); the bad-coordinate set B of Section 3; b_3, B_4 cubic and quartic coefficients in Section 7.
b, d, e_r	in Section 8, $b = \log 3$, d and e_r of (8.1). Elsewhere $d = (x - y)/2$ (Section 3), $d(u)$ the torus distance, d_3 a cubic coefficient, and $D(q 1/3)$ the Bernoulli relative entropy; D is also a Taylor degree in Section 6.
c, c_n	$c_n = \mu_n - 13/84$ is the Gaussian centre of the common score; $c = na$ is the scaled common Fourier variable in Section 4; c, c', c'' are also positive constants. The centre c_n is not the conditional mean (Proposition 5.1).
K, K_0	$K_0 = e^{-7865/32928}$ is the local correction (4.4); $K = \log n$ is the Gaussian truncation level in Section 6; k is the common score.
Θ, Θ_r, θ	the theta factor (1.2) and its residue values $\Theta_r = \Theta(c_n)$; $\theta \in [-\pi, \pi]^n$ is the Fourier variable.
$\rho_n, \lambda_n, \alpha, \beta$	the covariance eigenvalues (2.3) and $\alpha(q) = 2v(q)$, $\beta(q) = q(1 - q)$; $v(q)$ differs from the covariance entry v of (6.10), as u there differs from the one-match variable U .
M, M_j	$M = n(n - 1)$ is the number of matches; $M_j = \mathbb{E}f_D^j$ are moments in (6.12).
L, L_r	$L = \log y$ is the inverse target in Section 8; L_r are the degree- r logarithmic polynomials of Sections 6–7.
H_j, H_3	the periodic coefficient functions of Theorem 1.2; $H_3(z) = z^3 - 3z$ is a Hermite polynomial.
x, h	$x = k - c_n$ in Section 6; $x = \sqrt{L/b}$ and h in Section 8.

2 Mean matching and covariance

Tilt a single match by $e^{t \cdot \text{total score}}$. Its normalizer and draw probability are

$$A(t) = \frac{2e^{3t} + e^{2t}}{3}, \quad q = q(t) = \frac{1}{1 + 2e^t}.$$

The decisive outcomes have probability $(1 - q)/2$ each. Under this law, denoted $\mathbb{P}_q$, each team has mean $\mu(q) = (n - 1)(3 - q)$. For an interior target integer k , set $q = 3 - k/(n - 1)$; its tilted mean is exactly k . The change of measure is

$$\mathbb{P}(X = k\mathbf{1}) = e^{M \log A(t) - nkt} \mathbb{P}_q(X = k\mathbf{1}). \quad (2.1)$$

Here the unadorned $\mathbb{P}$ always means the original uniform law $q = 1/3$.

Put

$$v(q) = \frac{9 - 8q - q^2}{4}, \quad \beta(q) = q(1 - q), \quad \alpha(q) = 2v(q).$$

The covariance matrix has the form

$$\Sigma_{n,q} = \lambda_n(q)I + \frac{\rho_n(q) - \lambda_n(q)}{n} \mathbf{1}\mathbf{1}^T, \quad (2.2)$$

where its transverse and common eigenvalues are

$$\lambda_n(q) = n\alpha(q) - \beta(q), \quad \rho_n(q) = (n - 1)\beta(q). \quad (2.3)$$

Thus the normalized Gaussian integral is

$$J_n(q) = (2\pi)^{-n/2} \lambda_n(q)^{-(n-1)/2} \rho_n(q)^{-1/2}. \quad (2.4)$$

At $q_0 = 1/3$,

$$\lambda_n = \frac{28n - 2}{9}, \quad \rho_n = \frac{2(n - 1)}{9}, \quad \alpha = \frac{28}{9}, \quad \beta = \frac{2}{9}. \quad (2.5)$$

Both eigenvalues have order n . Using the full score vector at a fixed common score therefore avoids a singular covariance.

3 Sharp global localization

Let Q_0 be a sufficiently small fixed closed interval about $1/3$. All constants in this section are uniform for $q \in Q_0$. Denote by $\chi_{n,q}(\theta)$ the characteristic function of the centered total score vector. At a mean-matching integer score, Fourier inversion gives

$$\mathbb{P}_q(X = k\mathbf{1}) = (2\pi)^{-n} \int_{[-\pi, \pi]^n} \chi_{n,q}(\theta) d\theta.$$

Lemma 3.1 (Relative localization). *There is $c > 0$ such that*

$$(2\pi)^{-n} \int_{[-\pi, \pi]^n} |\chi_{n,q}(\theta)| d\theta \leq (1 + O(e^{-cn})) J_n(q). \quad (3.1)$$

For every fixed $A > 0$, the part of this integral outside $\max_i |\theta_i| \leq n^{-1/2} \log n$ is $o(n^{-A}) J_n(q)$.

3.1 Coercivity on the torus

The one-match kernel is

$$\phi_q(x, y) = \frac{1-q}{2}e^{3ix} + qe^{i(x+y)} + \frac{1-q}{2}e^{3iy}.$$

Let $d(u) = \min_{m \in \mathbb{Z}} |u - 2\pi m|$. The three probabilities are bounded below on Q_0 . Expanding the squared modulus and using $1 - \cos u \geq c d(u)^2$ gives

$$|\phi_q(x, y)|^2 \leq \exp\{-c[d(3x - 3y)^2 + d(2x - y)^2 + d(x - 2y)^2]\}. \quad (3.2)$$

For distinct i, j, k , the identity

$$\theta_j = (2\theta_i - \theta_j) + (2\theta_j - \theta_k) - (2\theta_i - \theta_k) \pmod{2\pi}$$

and the squared triangle inequality imply, after summation over ordered triples,

$$\sum_{i \neq j} d(2\theta_i - \theta_j)^2 \geq \frac{n-1}{9} \sum_j d(\theta_j)^2.$$

Indeed each of the three edge terms is counted $n-2$ times, and each left-side coordinate is counted $(n-1)(n-2)$ times before cancellation. Since the double-match modulus is $\prod_{i < j} |\phi_q(\theta_i, \theta_j)|^2$,

$$|\chi_{n,q}(\theta)| \leq e^{-cn \sum_i \theta_i^2} \quad (\theta \in [-\pi, \pi]^n, \ n \geq 3). \quad (3.3)$$

In particular the origin is the only modulus-one saddle. This coarse bound alone would lose a factor exponential in n relative to J_n ; the next two steps are essential.

3.2 A negative quartic and an exact Gaussian majorant

Set $s = (x + y)/2$ and $d = (x - y)/2$. The centered one-match phase is $sU + dV$, where (U, V) takes the values $(q, 3), (q, -3), (q - 1, 0)$ with probabilities $(1 - q)/2, (1 - q)/2, q$. Its fourth cumulant is

$$\begin{aligned} \kappa_4(sU + dV) &= q(1 - q)(1 - 6q + 6q^2)s^4 \\ &\quad + 54q(1 - q)(2q - 1)s^2d^2 + 81(1 - q)(3q - 2)d^4. \end{aligned} \quad (3.4)$$

At $q = 1/3$ this equals $-2s^4/27 - 4s^2d^2 - 54d^4$. All three coefficients stay strictly negative on a sufficiently small Q_0 . Uniform analyticity near the origin and evenness under $(s, d) \mapsto (-s, -d)$ give

$$\log |\phi_q| = -\frac{\kappa_2}{2} + \frac{\kappa_4}{24} + O((s^2 + d^2)^3).$$

Choose a fixed $\eta > 0$ so that the negative quartic dominates the remainder when $|x|, |y| \leq \eta$. Multiplying over all matches yields the sharp bound

$$|\chi_{n,q}(\theta)| \leq e^{-\theta^T \Sigma_{n,q} \theta / 2} \quad (\max_i |\theta_i| \leq \eta). \quad (3.5)$$

Its normalized integral is at most $J_n(q)$, with no dimension-dependent loss.

3.3 Outliers outside the fixed box

Fix $\varepsilon \in (0, 1/4)$. If at least εn coordinates have $|\theta_i| > \eta/10$, (3.3) gives e^{-cn^2} for the integrand. Its full contribution is negligible compared with $J_n(q) = \exp\{-\frac{n}{2} \log n + O(n)\}$.

Otherwise at least $(1 - \varepsilon)n$ coordinates are tiny, meaning $|\theta_i| \leq \eta/10$. Let $B = \{i : |\theta_i| > \eta\}$, with $m = |B| \geq 1$, and $K = B^c$, with $k = n - m \geq (1 - \varepsilon)n$. For a bad coordinate x and a tiny coordinate y ,

$$d(x - 2y) \geq d(x) - 2|y| \geq \frac{4}{5}|x|.$$

Their distinct edges give a factor $e^{-cn \sum_{i \in B} \theta_i^2}$. Retain every K - K edge and apply (3.5) to the complete k -team system; bound every remaining factor by 1. For a fixed set B , the unnormalized integral is consequently at most

$$I_k(q) \left[C n^{-1/2} e^{-c' n \eta^2} \right]^m, \quad I_k(q) = (2\pi)^k J_k(q). \quad (3.6)$$

This estimate is valid on the region with enough tiny coordinates; after the pointwise estimate that restriction can be removed for integration. There is no extra sum over choices of tiny coordinates.

For completeness, (2.3) gives the uniform comparison

$$I_k(q) \asymp [2\pi/\alpha(q)]^{k/2} k^{-k/2}$$

for all large k and all $q \in Q_0$. The residual factors are a bounded multiple of $(1 - \beta/(\alpha k))^{-(k-1)/2} \sqrt{k/(k-1)}$. It follows that

$$\frac{I_k(q) n^{-m/2}}{I_n(q)} \leq C D^m (n/k)^{k/2} \leq C (D e^{1/2})^m.$$

Summing (3.6) over all bad sets gives a relative bound

$$C \sum_{m \geq 1} \binom{n}{m} D_1^m e^{-c' n m} = O(e^{-c'' n}).$$

This proves exponentially small relative mass outside the fixed box.

3.4 Shrinking to the Gaussian scale

In the fixed box use (3.5). Every coordinate of $N(0, \Sigma_{n,q}^{-1})$ has variance $O(1/n)$. A union bound gives

$$\mathbb{P}\{\max_i |\theta_i| > n^{-1/2} \log n\} \leq 2n e^{-c \log^2 n} = o(n^{-A})$$

for every fixed A . Together with the outlier bound this proves Lemma 3.1.

4 The local integral and the score shift

We first work at $q = 1/3$. Under the Gaussian measure associated with J_n , decompose $\theta_i = y_i + a$, where $\sum_i y_i = 0$. The vector y and scalar a are independent, and may be represented as

$$y_i = \frac{Z_i - \bar{Z}}{\sqrt{\lambda_n}}, \quad a = \frac{W}{\sqrt{n\rho_n}}, \quad c = na,$$

with independent standard normals $Z_1, \dots, Z_n, W$. Put $Y_j = \sum_i y_i^j$. The cubic part of the double-match logarithm is exactly

$$-i \left[\frac{20n+1}{81} Y_3 + \frac{13n+1}{27} a Y_2 - \frac{n(n-1)}{81} a^3 \right]. \quad (4.1)$$

The quartic part at $a = 0$ is

$$-\frac{98n-1}{324} Y_4 - \frac{89}{108} Y_2^2. \quad (4.2)$$

The quartic terms involving a tend to zero in probability. On the shrinking box, the sum of degree at least five is $O(n^2(n^{-1/2} \log n)^5) = o(1)$.

The law of large numbers and central limit theorem give jointly

$$Y_2 \longrightarrow \frac{1}{\alpha}, \quad nY_4 \longrightarrow \frac{3}{\alpha^2}, \quad nY_3 \Rightarrow N(0, 6/\alpha^3), \quad c \Rightarrow N(0, 1/\beta). \quad (4.3)$$

The last variable is independent of the others. To verify the cubic limit, write $H_3(z) = z^3 - 3z$ and use

$$\sum_i (Z_i - \bar{Z})^3 = \sum_i H_3(Z_i) - 3\bar{Z} \left(\sum_i Z_i^2 - n \right) + 2n\bar{Z}^3.$$

After division by $\sqrt{n}$, the last two terms vanish in probability, whereas $\text{Var}(H_3(Z)) = 6$.

The quartic exponent converges to $-561/3136$. The transverse cubic has limiting variance $25/2058$, and the common cubic converges to $-i(13/84)c$. Their Gaussian expectations therefore give the correction

$$K_0 = \exp \left(-\frac{561}{3136} - \frac{25}{4116} - \frac{169}{3136} \right) = e^{-7865/32928}. \quad (4.4)$$

Passing to expectations is justified rather than formal: by (3.5), the complete correction relative to the Gaussian density has modulus at most 1 in the fixed box; its shrinking-box complement has superpolynomially small Gaussian measure. Bounded convergence in distribution applies to the displayed limiting correction. The same reasoning holds for every sequence $q \rightarrow 1/3$.

Now set $\delta = k - \mu_n$. For bounded δ ,

$$q = \frac{1}{3} - \frac{\delta}{n-1}, \quad t = \frac{9}{2} \frac{\delta}{n-1} + O(n^{-2}),$$

and Taylor expansion gives

$$M \log A(t) - nkt = -\frac{9}{4} \delta^2 + o(1), \quad (4.5)$$

$$\frac{J_n(q)}{J_n(1/3)} = \exp \left(-\frac{39}{56} \delta + o(1) \right). \quad (4.6)$$

For the determinant, note $\lambda'_n(1/3) = -(13n+1)/3$. Its $n-1$ transverse factors produce the finite linear term; the common eigenvalue changes only by relative $O(1/n)$. Combining (4.4)–(4.6) and completing the square proves

$$\mathbb{P}(X = k\mathbf{1}) = (1 + o(1)) J_n(1/3) e^{-12181/65856} e^{-9(\delta+13/84)^2/4}. \quad (4.7)$$

The error is uniform for bounded δ : otherwise a violating sequence has a convergent subsequence of δ and the preceding sequential argument gives a contradiction.

5 Dominated summation and the conditional common score

Pointwise fixed-score equivalents cannot be summed without a uniform tail estimate. If $|\delta| \leq \varepsilon_0 n$, with ε_0 sufficiently small, the exact mean-matching tilt lies in Q_0 . Lemma 3.1 implies $\mathbb{P}_q(X = k\mathbf{1}) \leq C J_n(q)$. The exact rate and determinant satisfy

$$M \log A(t) - nkt \leq -c\delta^2, \quad \frac{J_n(q)}{J_n(1/3)} \leq C e^{C|\delta|}.$$

For the first bound, $(\log A)'' = q(1 - q)$ is bounded above and below on the tilt interval; the second follows from the explicit eigenvalues. Thus

$$\frac{\mathbb{P}(X = k\mathbf{1})}{J_n(1/3)} \leq C e^{-c\delta^2 + C|\delta|}. \quad (5.1)$$

If $|\delta| > \varepsilon_0 n$, the tied event forces the total score nk to differ from its mean by at least $\varepsilon_0 n^2$. The number of draws is Binomial($M, 1/3$) and the total score is $3M$ minus this number. A binomial Chernoff bound makes the sum of these terms at most $O(n)e^{-cn^2} = o(J_n(1/3))$.

Dominated summation on each residue subsequence now gives

$$\mathbb{P}(\text{all tied}) \sim J_n(1/3) e^{-12181/65856} \sum_{k \in \mathbb{Z}} e^{-9(k-c_n)^2/4}. \quad (5.2)$$

The natural endpoint scores are in the negligible tail, so replacing the finite score range by $\mathbb{Z}$ loses no mass at this scale. Equations (1.3), (2.5), and (5.2) give (1.6). Finally,

$$\left(\frac{28n-2}{28n} \right)^{-(n-1)/2} \rightarrow e^{1/28}, \quad \sqrt{\frac{n}{n-1}} \rightarrow 1,$$

and $1/28 - 12181/65856 = -9829/65856$, proving Theorem 1.1.

Proposition 5.1 (Conditional score law). *On each residue subsequence, after an integer translation, the common score conditioned on all teams being tied converges to the discrete Gaussian with weights proportional to $e^{-9(k-c_n)^2/4}$. Every fixed moment converges as well. In particular,*

$$\mathbb{E}[k \mid \text{all tied}] - \mu_n = -\frac{13}{84} + \frac{2}{9} \frac{\Theta'(c_n)}{\Theta(c_n)} + o(1), \quad (5.3)$$

$$\text{Var}(k \mid \text{all tied}) = \frac{2}{9} + \frac{4}{81} (\log \Theta)''(c_n) + o(1). \quad (5.4)$$

Proof. The same majorant (5.1) permits multiplication by every fixed power of $|\delta|$. If $Z(c) = \sum_k e^{-9(k-c)^2/4}$, differentiation gives $\mathbb{E}(k - c) = \frac{2}{9} (\log Z)'$ and $\text{Var}(k) = \frac{2}{9} + \frac{4}{81} (\log Z)''$. Use $Z(c) = 2\sqrt{\pi}\Theta(c)/3$. $\square$

The parameter c_n is a Gaussian center, generally not the actual conditional mean. The shift $-13/84$ comes from the mixed common-transverse cubic; omitting it changes the leading theta multiplier. On the tied event, the number of draws is exactly $n[3(n-1) - k]$ and hence is a multiple of n .

6 Expansion to every fixed order

We now establish a uniform local expansion for $q \in Q_0$, then perform the tilted score sum. The external input is Isaev's complex cumulant theorem [3, Theorem 1.2]. We state the form used here to make the hypothesis verification explicit.

6.1 The complex cumulant input

Let $Z_1, \dots, Z_n$ be independent on a product domain Ω and let $f : \Omega \rightarrow \mathbb{C}$ be bounded. For coordinate replacement $z \triangleright_W y$, define

$$\bar{\Delta}_V(f) = \sup_{y \in \Omega} \left| \mathbb{E} \sum_{W \subseteq V} (-1)^{|W|} f(Z \triangleright_W y) \right|, \quad S_k(f) = \max_j \sum_{\substack{|V|=k \\ j \in V}} \bar{\Delta}_V(f).$$

For a fixed integer $m \geq 1$, suppose $0 \leq \gamma \leq 1/100$ and

$$\sum_{k=t}^n e^{3\gamma k/2} 2^t \binom{k-1}{t-1} S_k(f) \leq \gamma \quad (1 \leq t \leq m). \quad (6.1)$$

Then

$$\mathbb{E} e^{f(Z)} = (1 + \zeta)^n \exp \left(\sum_{j=1}^m \frac{\kappa_j(f(Z))}{j!} \right), \quad |\zeta| \leq e^{(100\gamma)^{m+1}} - 1. \quad (6.2)$$

For $2 \leq j \leq m$ the theorem also gives

$$|\kappa_j(f(Z))| \leq \frac{n(j-1)!(80\gamma)^j}{50j}.$$

These are algebraic complex cumulants without conjugation. Bounding each $\bar{\Delta}_V$ by its supremum mixed difference is sufficient for (6.1).

6.2 Symmetric whitening and independent truncation

The symmetric inverse square root of (2.2) is

$$T = \lambda^{-1/2} I + \frac{\rho^{-1/2} - \lambda^{-1/2}}{n} \mathbf{1}\mathbf{1}^T, \quad \theta = TZ, \quad (6.3)$$

with independent standard normals Z_i . Diagonal entries of T are $O(n^{-1/2})$, off-diagonal entries are $O(n^{-3/2})$, and

$$\max_i \sum_j |T_{ij}| + \max_j \sum_i |T_{ij}| \leq Cn^{-1/2} \quad (6.4)$$

uniformly in q . This spreads the common fluctuation among the n independent variables; it does not introduce a single coordinate with order-one influence in the cubic exponent.

Take $K = \log n$ and condition each Z_i on $|Z_i| \leq K$. Independence is preserved. The omitted probability is smaller than every inverse power of n , and (6.4) puts the image of the product cube inside $\max_i |\theta_i| \leq CK/\sqrt{n} < \eta$. Conversely, removing from the fixed θ -box the part outside this image costs at most the omitted Gaussian probability, by (3.5). The normalization introduced by conditioning also differs from 1 by a superpolynomially small amount.

Let $L_r(\theta)$ be the degree- r term in the logarithm of the centered characteristic function, with the quadratic removed, and put

$$f_D(Z) = \sum_{r=3}^D L_r(TZ).$$

Each L_r is a sum over edges of bounded-coefficient homogeneous bivariate polynomials. On the product cube its Taylor error is

$$O_D(n^2(K/\sqrt{n})^{D+1}) = O_D(n^{(3-D)/2} K^{D+1}). \quad (6.5)$$

The exact correction has modulus at most 1, and the truncated correction has modulus at most the exponential of this error. Hence replacement of the exponent changes the normalized integral by the same order.

6.3 Mixed derivative bounds

For $1 \leq k \leq r$ and a fixed index j_1 , the chain rule gives

$$\begin{aligned} & \sum_{j_2, \dots, j_k} \sup_{|Z_i| \leq K} |\partial_{j_1} \cdots \partial_{j_k} L_r(TZ)| \\ & \leq C_r (K/\sqrt{n})^{r-k} \left[\sum_{uv \text{ edge}} (|T_{u,j_1}| + |T_{v,j_1}|) \right] \left[\max_{uv \text{ edge}} \sum_j (|T_{u,j}| + |T_{v,j}|) \right]^{k-1} \\ & \leq C_r n^{1-r/2} K^{r-k}. \end{aligned} \quad (6.6)$$

The first bracket is $(n-1)$ times a column absolute sum, hence $O(\sqrt{n})$; the second is $O(n^{-1/2})$. Including repeated indices only enlarges the bound, so it controls the subset sums required above. A k -fold mixed difference on the cube is bounded by $(2K)^k$ times its mixed derivative. Consequently

$$S_k(f_D) \leq C_D n^{-1/2} K^D \quad (1 \leq k \leq D), \quad S_k(f_D) = 0 \quad (k > D). \quad (6.7)$$

For fixed D, m , choose $\gamma = C_{D,m} n^{-1/2} K^D$ with a sufficiently large constant. There are only finitely many nonzero terms in (6.1); it holds for all large n , as does $\gamma \leq 1/100$. The relative error in (6.2) is therefore

$$O_{D,m}(n\gamma^{m+1}) = O_{D,m}(n^{(1-m)/2} K^{D(m+1)}). \quad (6.8)$$

Choosing fixed D, m sufficiently large gives any requested inverse-power precision. In particular, this resums the order-one cubic fluctuation; no finite Taylor expansion of its exponential is being presumed valid.

Truncated-normal cumulants can be replaced by full-normal cumulants with superpolynomial error. Every fixed moment of f_D is a fixed-degree Gaussian polynomial with coefficients growing at most polynomially in n . Cauchy–Schwarz and Gaussian polynomial moment bounds make its tail contribution at most $n^C e^{-c \log^2 n}$. Cumulants are finite polynomial combinations of moments. Thus, to any prescribed inverse-power precision, the local correction equals

$$\exp \left(\sum_{j=1}^m \frac{\kappa_j(f_D(Z))}{j!} \right) (1 + \text{controlled error}), \quad (6.9)$$

where now Z is an unconditioned standard Gaussian vector. The leading limit is positive uniformly on a sufficiently small Q_0 , so the additive Taylor and localization errors can also be expressed relatively.

6.4 Exact finite coefficients and integer powers

For clarity use u, v for the two covariance entries:

$$u = \lambda^{-1}, \quad v = \frac{\rho^{-1} - \lambda^{-1}}{n}, \quad \mathbb{E} \theta_i \theta_j = u \mathbf{1}_{i=j} + v. \quad (6.10)$$

They are rational functions of n, q . For fixed degree, every moment of products of the L_r is rational in n, q by Wick's formula: grouping vertex identifications by a finite set partition yields a falling factorial $(n)_\ell$ times a finite Gaussian pairing sum. Here is a completely explicit rule. Let $P_r = \sum_i \theta_i^r$, $P_0 = n$. For a product $P_{r_1} \cdots P_{r_t}$, sum over all set partitions π of

$\{1, \dots, t\}$. Put $d_B = \sum_{j \in B} r_j$ for each block B , $d = \sum_B d_B$, and $h = \sum_B h_B$. If d is odd the moment is zero. If d is even the contribution of π is

$$(n)_{|\pi|} \sum_{0 \leq h_B \leq \lfloor d_B/2 \rfloor} \left[\prod_{B \in \pi} \binom{d_B}{2h_B} (2h_B - 1)!! \right] (d - 2h - 1)!! u^h v^{d/2-h}. \quad (6.11)$$

Here $(-1)!! = 1$. To prove the rule, for distinct block indices represent $\theta_B = \sqrt{u} Z_B + \sqrt{v} W$ and expand, using Gaussian even moments. On Q_0 , $v > 0$; the resulting polynomial identity is algebraic regardless of this representation. Factors P_0 can simply be taken out as powers of n . Every symmetric bivariate edge polynomial is a finite combination of the P_r , since

$$\sum_{i \neq j} \theta_i^a \theta_j^b = P_a P_b - P_{a+b}.$$

If $M_j = \mathbb{E} f_D^j$, with $M_0 = 1$, the ordinary cumulant recursion is

$$\kappa_j = M_j - \sum_{i=1}^{j-1} \binom{j-1}{i-1} \kappa_i M_{j-i}. \quad (6.12)$$

Equations (6.11)–(6.12) give a finite rational algorithm. The first cumulant is $O(1)$: odd-degree means vanish and even $r \geq 4$ have mean $O(n^{2-r/2})$. Isaev's bound controls higher cumulants by $O(n^{1-j/2} \log^C n)$. Since they are rational functions with integer-power Laurent expansions at infinity, no positive powers can occur, and no constant term can occur for $j \geq 3$. Thus the logarithm in (6.9) has an ordinary $1/n$ expansion with coefficients regular uniformly on Q_0 .

For a requested grade J , a finite procedure is:

1. Choose $D \geq 2J + 8$ and $m \geq 2J + 5$. Both (6.5) and (6.8) are $o(n^{-J-1})$.
2. Compute the one-match cumulants through degree D from the three-point distribution, and sum the double-match logarithmic terms.
3. Use (6.11) and (6.12) to form $\sum_{j \leq m} \kappa_j(f_D)/j!$ exactly.
4. Expand this rational function through n^{-J-1} and exponentiate as a finite formal series. Retaining grades through J leaves $O_J(n^{-J-1})$.

Increasing D, m cannot change the resulting coefficients, by uniqueness of asymptotic power series for the same local integral.

6.5 From local series to theta derivatives

Substitute $q = 1/3 - \delta/(n-1)$ into the uniform expansion and multiply by the exact tilt factor and determinant. Each coefficient is a polynomial in δ times $e^{-9(\delta+13/84)^2/4}$. On $|\delta| \leq \log n$, Taylor remainders are bounded by n^{-J-1} times a fixed polynomial in $|\delta|$ times $e^{-c\delta^2+C|\delta|}$. The sum of this bound is uniform; the part $|\delta| > \log n$ is superpolynomially negligible by (5.1) and the large-deviation bound. This proves the remainder in (1.7).

Finally put $x = k - c_n$, so $\delta = x - 13/84$. Differentiating $Z(c) = \sum_k e^{-9(k-c)^2/4}$ expresses every Gaussian power sum as a rational linear combination of derivatives of Z : for example, $Z' = \frac{9}{2} \sum_k (k-c) e^{-9(k-c)^2/4}$. Recursing through higher derivatives yields the claimed structure of H_j in Theorem 1.2. On the three residue subsequences, these functions take constant values.

7 The first correction

The preceding argument proves existence and error control at every fixed order. We now give the finite algebra for its first nonconstant term. At $q = 1/3$, the necessary logarithmic polynomials are

$$L_3 = i \left(\frac{7}{27} P_1 P_2 - \frac{20n+1}{81} P_3 \right), \quad (7.1)$$

$$L_4 = \frac{91}{81} P_1 P_3 - \frac{89}{108} P_2^2 - \frac{98n-1}{324} P_4, \quad (7.2)$$

$$L_5 = i \left(\frac{407}{972} P_1 P_4 - \frac{133}{486} P_2 P_3 - \frac{140n+1}{972} P_5 \right), \quad (7.3)$$

$$L_6 = \frac{7861}{14580} P_1 P_5 - \frac{8015}{5832} P_2 P_4 + \frac{8077}{8748} P_3^2 - \frac{7718n-7}{87480} P_6. \quad (7.4)$$

Take $D = 6, m = 4$. The Taylor and cumulant-tail errors are $O(n^{-3/2} \log^C n) = o(1/n)$. Through grade $1/n$, the logarithmic correction consists of

$$\begin{aligned} \mathbb{E}L_4 + \frac{1}{2}\mathbb{E}L_3^2 + \mathbb{E}L_6 + \mathbb{E}(L_3 L_5) + \frac{1}{2}\text{Var}(L_4) \\ + \frac{1}{2}\kappa(L_3, L_3, L_4) + \frac{1}{24}\kappa_4(L_3). \end{aligned} \quad (7.5)$$

All variances and cumulants here are algebraic, without complex conjugation.

7.1 Why the omitted mixtures are smaller

In every fixed Gaussian L^p norm,

$$L_3 = O(1), \quad L_4 - \mathbb{E}L_4 = O(n^{-1/2}), \quad L_5 = O(n^{-1}), \quad L_6 - \mathbb{E}L_6 = O(n^{-3/2}).$$

These assertions follow by inserting $\theta_i = y_i + c/n$ into (7.1)–(7.4) and using Gaussian polynomial moment bounds. More specifically,

$$(L_3, \sqrt{n}(L_4 - \mathbb{E}L_4), nL_5)$$

converges jointly, with all fixed moments, to a centered Gaussian vector. The first and third coordinates are linear combinations of the common normal c and empirical Hermite sums of degrees 3 and 5; the second is a linear combination of Hermite sums of degrees 2 and 4. The even component is independent of the odd components in this Gaussian limit. Also nL_6 tends to a deterministic constant.

Consequently the potentially order- $1/n$ mixtures $\kappa(L_3, L_3, L_4, L_4)$ and $\kappa(L_3, L_3, L_3, L_5)$ are actually $o(1/n)$: their scaled limits are fourth cumulants of a Gaussian vector. Parity kills the odd total-degree terms exactly. The displayed L^p estimates control the other omissions; cumulants of order above four are covered by (6.8). This establishes (7.5) without supposing $L_3 = o(1)$.

7.2 Exact local coefficient

The finite Wick calculation gives the following constants and $1/n$ coefficients. The entries are exact rational numbers.

Quantity	Constant	Coefficient of $1/n$
$\mathbb{E}L_4$	$-561/3136$	$159/5488$
$\mathbb{E}L_6$	0	$-31763/175616$
$\mathbb{E}L_3^2$	$-3949/32928$	$14407/115248$
$\mathbb{E}(L_3L_5)$	0	$-26561/175616$
$\text{Var}(L_4)$	0	$343533/1229312$
$\kappa(L_3, L_3, L_4)$	0	$492179/2151296$
$\kappa_4(L_3)$	0	$4310925/15059072$

Thus

$$\log(\text{local correction at } q = 1/3) = -\frac{7865}{32928} + \frac{1147975}{45177216} \frac{1}{n} + O(n^{-2}). \quad (7.6)$$

The initial calculation gives $o(1/n)$; the already proved all-order expansion upgrades this to $O(n^{-2})$. A second raw-moment aggregation through cumulant order six yields the same two coefficients and zero constant and $1/n$ terms for the fifth and sixth cumulants. A further Gaussian integration-by-parts computation uses the coordinate covariance directly and reproduces these first-correction moments and the vanishing coefficients of the two omitted fourth-cumulant mixtures.

7.3 Tilt and carrier contributions

For general q , put

$$d_3 = \frac{\beta(q)(q+13)}{12}, \quad b_3 = \frac{\beta(q)(q+4)}{2},$$

and let A_4, B_4, C_4 be the three quartic coefficients in (3.4). The leading local logarithmic correction is

$$\begin{aligned} C(q) &= \frac{3A_4 + B_4 + 3C_4}{96\alpha(q)^2} - \frac{3d_3^2}{\alpha(q)^3} - \frac{b_3^2}{2\beta(q)\alpha(q)^2} \\ &= -\frac{q^4 - 28q^3 - 122q^2 + 1944q - 2187}{12(q-1)(q+9)^3}. \end{aligned} \quad (7.7)$$

In particular $C(1/3) = -7865/32928$ and $C'(1/3) = 531/153664$. The dependence of the leading local correction on the tilt contributes $-(531/153664)\delta/n$.

The exact tilt rate is $-n(n-1)D(q||1/3)$, where D is Bernoulli relative entropy. Its expansion and that of the determinant are

$$-n(n-1)D(q||1/3) = -\frac{9}{4}\delta^2 + \frac{1}{n} \left(-\frac{9}{4}\delta^2 - \frac{9}{8}\delta^3 \right) + O(n^{-2} \text{ poly}(\delta)), \quad (7.8)$$

$$\log \frac{J_n(q)}{J_n(1/3)} = -\frac{39}{56}\delta + \frac{1}{n} \left(\frac{507}{784}\delta + \frac{1773}{3136}\delta^2 \right) + O(n^{-2} \text{ poly}(\delta)). \quad (7.9)$$

Combining these with (7.6) gives, relative to the exact $J_n(1/3)$ shifted-Gaussian carrier, the first correction

$$\frac{1147975}{45177216} + \frac{98841}{153664}\delta - \frac{5283}{3136}\delta^2 - \frac{9}{8}\delta^3.$$

The logarithm of the remaining covariance normalization factor is

$$\log \left[\sqrt{\frac{n}{n-1}} \left(1 - \frac{1}{14n} \right)^{-(n-1)/2} \right] = \frac{1}{28} + \frac{365}{784n} + O(n^{-2}).$$

The constant $1/28$ is already included in B_n . Adding $365/784$ to the displayed polynomial's constant gives (1.8). Dominated summation proves (1.9)–(1.10).

8 Residue-aware asymptotic inversion

Let $b = \log 3$ and set

$$d = -b + \frac{1}{2} \log \frac{9}{56\pi}, \quad e_r = -\frac{1}{2} \log \frac{9}{56\pi} - \frac{9829}{65856} + \log \Theta_r, \quad (8.1)$$

where $\Theta_r = \Theta(c_n)$ for $n \equiv r \pmod{3}$. Taking logarithms of Theorem 1.2,

$$\log a(n) = bn^2 - \frac{n}{2} \log n + dn + e_r + \sum_{j=1}^R \ell_{j,r} n^{-j} + O_R(n^{-R-1}), \quad \ell_{1,r} = R_r. \quad (8.2)$$

There is no standalone $\log n$ term: the two logarithms from $n^{-1/2}$ and $n^{-(n-1)/2}$ cancel it exactly.

For a fixed R , define the explicit smooth function

$$F_{R,r}(t) = bt^2 - \frac{t}{2} \log t + dt + e_r + \sum_{j=1}^R \ell_{j,r} t^{-j}. \quad (8.3)$$

Its derivative is $2bt + O(\log t) > 0$ for large t , so its large real inverse $\nu_{R,r}(L)$ exists uniquely. This differentiates the explicit approximant, not an unspecified asymptotic remainder. For example, put

$$x = \sqrt{L/b}, \quad h = \frac{\log x}{4b} - \frac{d}{2b}.$$

The first three terms of the smooth inverse are

$$\nu_{0,r}(L) = x + h + \frac{-bh^2 + \frac{h}{2}(\log x + 1) - dh - e_r}{2bx} + o(x^{-1}). \quad (8.4)$$

Successive substitution or Newton coefficient matching produces any fixed further grade in x^{-1} , with polynomial coefficients in $\log x$. The derivative bound for $F_{R,r}$ controls the remainder.

Proposition 8.1 (Integer threshold enclosure). *For fixed R and all sufficiently large L , let $\nu_r = \nu_{R,r}(L)$ and $\epsilon_r = C\nu_r^{-R-2}$, with C sufficiently large. Define*

$$N_r^- = r + 3 \left\lceil \frac{\nu_r - \epsilon_r - r}{3} \right\rceil, \quad N_r^+ = r + 3 \left\lceil \frac{\nu_r + \epsilon_r - r}{3} \right\rceil.$$

If $N(L) = \min\{n \geq 1 : a(n) \geq e^L\}$, then

$$\min_{r=0,1,2} N_r^- \leq N(L) \leq \min_{r=0,1,2} N_r^+. \quad (8.5)$$

If a truncated inverse series replaces ν_r , its independently bounded absolute error must be added to ϵ_r .

Proof. By (8.2), the forward error near ν_r is $O(\nu_r^{-R-1})$. The derivative of (8.3) is bounded below by a positive multiple of ν_r there. Hence values below $\nu_r - C\nu_r^{-R-2}$ cannot cross the threshold, and values above $\nu_r + C\nu_r^{-R-2}$ do cross it, with C large enough. Taking the next integer in residue class r proves the two bounds for that residue and then (8.5). Eventual strict monotonicity also follows directly from the integer forward formula: the successive main-term difference is $2bn + O(\log n)$, the residue constants are bounded, and the errors tend to zero. Finite initial exceptions do not affect large thresholds. $\square$

The asymptotic O -constants and their onset are not numerically certified here. These are eventual theoretical enclosures, not ready-made finite input certificates. If a branch lies within its uncertainty of the residue lattice, no floor-free asymptotic formula resolves the rounding.

[write] **Added 2 October 2026, batch 77: relation to repository results.** The inversion of this section falls under the general inversion apparatus of ProveIt’s transseries volumes [8, 9], and no novelty is claimed for the method. The dominant block here is quadratic, bt^2 , so the Lambert cores (`p0:thm:lambert-core`, `p0:prop:factorial-core`) and the linear-logarithmic reversion `p0:thm:lambert-centered` of [8] do not apply: the seed is $x = \sqrt{L/b}$ and no Lambert function occurs. The phase $bt^2 - \frac{t}{2} \log t + dt$ is an admissible core in the sense of `p0:def:core` of [8] (its slope tends to $+\infty$), and the successive substitution or Newton coefficient matching described after (8.4) is the formal reversion about such a core, `p0:thm:core-reversion` of [8] (in [9], the scaled local reversion `t2:prop:scaled`, stated for an arbitrary core). The residue-aware enclosure of Proposition 8.1 combines two general parts of `p0:thm:staircase`: part (4), the residue-class formula $\min_{\varrho}(\varrho + m\lceil(\nu_{\varrho} - \varrho)/m\rceil)$ (written there with modulus r and residue ρ ; here the modulus is $m = 3$, the residue is called r , and ρ_n is an eigenvalue), which is the shape of $N_r^{\pm}$ and (8.5); and the separation condition of part (2), applied in each residue class with the width ϵ_r , because $F_{R,r}$ is a smooth approximant and not an admissible interpolation of $a(n)$ on the class. The closing remark above is the failure mode of part (2). The arithmetic of those two parts is formalized in Lean as `Fabius.staircase_separation` and `Fabius.isLeast_residue_class` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; those statements concern arbitrary real numbers and monotone functions and give no formal status to anything in this section. Specific to this report are the forward expansion (8.2), with its three residue constants e_r and $\ell_{1,r} = R_r$ and the exact cancellation of the standalone $\log n$ term, and the explicit terms of (8.4).

9 Prior work and scope

The football model and its exact finite enumeration belong to Jehn, Habermann and Lavrov [1]; the present argument concerns its large- n asymptotics. A bounded search of the exact sequence, title, and related football-score and dense-graph asymptotic literature, completed on 2 October 2026, located no football-specific asymptotic for this count. That negative search is not an exhaustive novelty certificate.

Several substantial general precedents are relevant. Barvinok and Hartigan [4] give maximum-entropy Edgeworth estimates, with hypotheses that include local correction and global localization. Isaev and McKay [6] treat complex perturbed Gaussian integrals. Isaev, McKay and Zhang [5] develop cumulant expansions for Eulerian orientations; their real-valued theorem cannot simply be applied to the present complex football exponent. Isaev’s complex theorem [3] is the explicit higher-order input used in Section 6. The graph-factor work of Isaev and McKay [7] provides a close methodological precedent for whitening and derivative estimates, but its Bernoulli edge model does not immediately give the three-outcome football count.

Accordingly, no new general local-limit or cumulant theorem is claimed. The model-specific work consists of the sharp localization, the common score tilt and lattice summation, their constants and shift, the checked application of the complex theorem, and the resulting explicit first correction and inverse formulation.

10 Reproducibility and limits

The accompanying source package contains this editable LaTeX report, the compiled PDF, portable Python scripts, frozen expected JSON results, and a SHA256 manifest. From the extracted package root, install the listed dependencies and run `python replay.py`. Generated outputs are written separately from the expected fixtures. The package-local `build-report.sh` rebuilds the PDF with a standard TeX installation. The README gives exact commands and a map of the files.

The replay checks one-match cumulants, symmetric-polynomial identities, leading constants, exact counts through four teams, the finite Wick calculation for the first correction, an alternative raw-cumulant aggregation, a Gaussian integration-by-parts calculation, residue constants and conditional moments, and selected finite- n Gaussian-cumulant diagnostics. The known counts for five through eight teams are comparison data from [1, 2], not recomputed exact enumerations in this replay. The generic every-fixed-order algorithm is specified mathematically in (6.11)–(6.12); the supplied programs implement the coefficient checks stated above, rather than an optimized arbitrary-order symbolic engine.

Finite algebraic checks do not replace the localization, error bounds, or convergence arguments. The proof establishes every fixed order, not convergence of the infinite asymptotic series or uniformity when the truncation order grows with n . Numerical theta and correction values are diagnostic decimals; no interval-certified constants, effective remainder constants, or certified finite- n approximation thresholds are asserted. Small- n comparisons are not evidence of an error bound. Only original report and verification material is packaged; cited papers remain available at their primary source links.

[write] **Added 2 October 2026, batch 77.** The source package described above is not shipped as such. In the collection the nine programs, `replay.py` and `build-report.sh` are in `code/`, and the seven frozen fixtures `expected/<name>.json` and `requirements.txt` in `data/` (as `data/expected-<name>.json`), all byte-identical to the delivery; the compiled PDF and the SHA256 manifest `MANIFEST.sha256` (verified at intake and retired) are not shipped and survive in the archive. `replay.py` locates the checks under `code/` and the fixtures under `expected/` relative to its own directory, so in the collection layout it finds neither; `build-report.sh` compiles `report/tied-football-asymptotics.tex`, which is shipped only as this `article.tex`. The individual check scripts write their default output beside the report unless directed elsewhere. The report's `README.md` gives a replay on a scratch copy, rebuilt in the delivered layout from the shipped files, that leaves them untouched.

References

- [1] R. Jehn, K. Habermann, and M. Lavrov, *Number of ways that a football league can complete with all teams having the same number of points*, arXiv:2503.14509v1 (2025). <https://arxiv.org/abs/2503.14509>.
- [2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A380592. <https://oeis.org/A380592>.
- [3] M. Isaev, *A tail bound for cumulant series for complex functions of independent random variables*, arXiv:2508.16952v2 (2025), Theorem 1.2. <https://arxiv.org/abs/2508.16952v2>.
- [4] A. Barvinok and J. A. Hartigan, *Maximum entropy Edgeworth estimates of the number of integer points in polytopes*, arXiv:0910.2497v2, Theorem 1. <https://arxiv.org/abs/0910.2497>.
- [5] M. Isaev, B. D. McKay, and R.-R. Zhang, *Cumulant expansion for counting Eulerian orientations*, *Journal of Combinatorial Theory, Series B* **172** (2025), 263–314. <https://arxiv.org/abs/2309.15473v2>.
- [6] M. Isaev and B. D. McKay, *Complex martingales and asymptotic enumeration*, *Random Structures & Algorithms* **52** (2018), 617–661. <https://arxiv.org/abs/1604.08305>.
- [7] M. Isaev and B. D. McKay, *Asymptotic enumeration of graph factors by cumulant expansion*, arXiv:2508.18731v1 (2025). <https://arxiv.org/abs/2508.18731>.

- [8] [write] ProveIt, *Transseries and inversion*, the transseries-and-inversion volume (p0:def:core, p0:thm:core-reversion, p0:thm:staircase; not applicable: p0:thm:lambert-core, p0:prop:factorial-core, p0:thm:lambert-centered), Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex.
- [9] [write] ProveIt, *Combinatorial transseries inverses* (t2:prop:scaled), Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex.